RELATIONS

SYNOPSIS

- **Cartesian product of sets:** Let P and Q be two non empty sets. The set of all ordered pairs (a,b) such that $a \in P$ and $b \in Q$ is called the cartesian product of set P with set Q and is denoted by $P \times Q$.

  Thus $P \times Q = \{(a,b) : a \in P \text{ and } b \in Q\}$

  If P or Q is a null set then $P \times Q$ will also a null set.

  Note: (i) Two ordered pairs are equal , if and only if the corresponding first elements are equal and the second elements are also equal.
  (ii) If n(A) = p, n(B)=q ,then n(A x B)=pq

- $A \times A \times A = \{(a,b,c) : a,b,c \in A\}$. Here (a,b,c) is called an ordered triplet.

  Eg : If $A = \{1,2,3\}$ and $B = \{a,b\}$ then

  i) $A \times B = \{(1,a),(1,b),(2,a),(2,b),(3,a),(3,b)\}$

  ii) $B \times A = \{(a,1),(b,1),(a,2),(b,2),(a,3),(b,3)\}$

  iii) $A \times A = \{(1,1),(1,2),(1,3),(2,1),$
  $(2,2),(2,3),(3,1),(3,2),(3,3)\}$

  iv) $B \times B = \{(a,a),(a,b),(b,a),(b,b)\}$

- **Relation :** Let A and B be two non empty sets. Then any subset of $A \times B$ is called a relation R from A to B. Here A is called domain and B is called codomain of R

  The set of the all second elements in a relation R from a set A to set B is called the range of the relation R.

  Note: 1) Range $\subseteq$ Codomain
  2) A relation may be represented algebrically either by the roster method or by the set builder method.
  3) The total number of relations that can be defined from a set A to set B is the number of possible subsets of $A \times B$.

  If $n(A) = p$ and $n(B) = q$ then $n(A \times B) = pq$ and the total number of relations is 2^{pq}.

  Eg 1 : If $A = \{2,3,5,6\}$ and R be a relation "divides" on A such that a Rb $\Leftrightarrow$ a divides b

  $\therefore R = \{(2,2)(2,6)(3,3)(3,6)(5,5)(6,6)\}$

  Eg 2 : If $A = \{1,2,3\}, B = \{a,b,c\}$ and $R = \{(1,a)(1,c)(2,b)\}$ Then

  Domain of $R = \{1,2\}$, Range of $R = \{a,b,c\}$

  Eg 3 : Let R be a relation on the set N of natural numbers defined by $a + 3b = 12$ Then

  $R = \{(9,1)(6,2)(3,3)\}$

  $\therefore$ Domain of $R = \{9,6,3\}$ and Range of $R = \{1,2,3\}$

WE-1: Let $A = \{1,2,3,4,6\}$ let R be the relation on A defined by $\{(a,b) : a,b \in A,$ b is exactly divisible by a$\}$ a) write R in roster form b) find the domain of R c) find the range of R

Sol: a) $R = \{(1,1),(1,2),(1,3),(1,4),(1,6),(2,4),(2,6),$
$(2,2),(4,4),(6,6),(3,3),(3,6)\}$

b) Domain of $R = \{1,2,3,4,6\}$

c) Range of $R = \{1,2,3,4,6\}$

- **Inverse Relation:** Let A and B be two sets and let R be a relation from a set A to B. Then the inverse of R, denoted by R^{-1}, is a relation from B to A and is defined by

  $R^{-1} = \{(b,a);(a,b) \in R\}$.

  Also dom(R)=range (R^{-1}) and

  range (R)=dom (R^{-1})

  Eg : If $A = \{1,2,3\}$ and $B = \{a,b,c\}$

if $R=\{(1,a),(2,a),(3,b),(3,c)\}$

then $R^{-1}=\{(a,1),(a,2),(b,3),(c,3)\}$

➢ **Types of relations :**

Void Relation: The relation having no ordered pairs is called a void relation and is denoted by ϕ

Universal Relation: Let A be a set then $A\times A$ is called universal relation on A

Note: The void and the universal relations on a set A are respectively the smallest and the largest relations on A.

Eg : If $A=\{1,2\}$ then universal relation in A is $AXA=\{(1,1),(1,2),(2,1),(2,2)\}$

Identity Relation: Let A be a set. Then the relation $\{(a,a):a\in A\}$ is called the identity relation on A and is denoted by I_A.

Eg : If $A=\{1,2,3\}$ then

$I_A=\{(1,1),(2,2),(3,3)\}$ is the identity relation on A.

Reflexive Relation: A relation R on a set A is said to be reflexive if every element of A is related to itself. Thus R is reflexive $\Leftrightarrow (a,a)\in R\ \forall a\in A$

A relation R on a set A is not reflexive if there exists an element $a\in A$ such that $(a,a)\notin R$.

Note: Every identity relation is reflexive but every reflexive relation need not be an identity relation.

Eg : If $A=\{1,2,3\}$ and $R=\{(1,1),(2,2)\}$

then R is not reflexive since $3\in A$ but $(3,3)\notin R$

W.E-2: For real numbers x and y we write x Ry iff $x-y+\sqrt{2}$ is an irrational number. Then the relation R is Reflexive.

Sol: Given that in the set of real numbers R,

xRy iff $x-y+\sqrt{2}$ is irrational.

Reflexive: $x-x+\sqrt{2}=\sqrt{2}$ (irrational).

for every real x.

➢ **Symmetric relation :** A relation R on a set A is said to be a **symmetric relation** iff

$(a,b)\in R\Rightarrow(b,a)\in R$ for all $a,b\in A$

i.e $aRb\Rightarrow bRa$ for all $a,b\in A$

Eg : If $A=\{2,4,6,8\}$ and

$R=\{(2,4)(4,2)(4,6)(6,4)\}$ then R is symmetric.

➢ **Transitive Relation:** Let A be any set. A relation R on A is said to be transitive relation iff $(a,b)\in R$ and $(b,c)\in R\Rightarrow(a,c)\in R$ for all $a,b,c\in A$

i.e aRb and bRc $\Rightarrow$ aRc for all $a,b,c\in A$

W.E-3: A relation R defined on the set of integers $R=\{(a,b)$: a divides b; $a,b\in Z\}$ then R is______

Sol: R is not reflexive since $(0,0)\notin R$, since 0 does not divide 0.

R is not symmetric , since 2/4 but 4/2.

R is transitive since

a/b,b/c$\Rightarrow b=an, c=bm,\ n,m\in Z$

$\Rightarrow c=a(nm)\ \Rightarrow a/c.$

$\therefore R$ is neither reflexive, nor symmetric but only transitive.

➢ **Antisymmetric Relation :** Let A be any set. A relation R on set A is said to be antisymmetric relation iff $(a,b)\in R$ and $(b,a)\in R\Rightarrow a=b$ for all $a,b\in A$.

Note: If $(a,b)\in R$ and $(b,a)\notin R$ then still R is an anti symmetric relation

W.E-4: Let N be the set of natural numbers. A relation $R\subseteq N\times N$ is defined by 'x divides y' is anti symmetric

Sol: xRy , $yRx\Rightarrow x$ divides y, y divides x $\Rightarrow x=y$.

➢ **Equivalence Relation:** A relation R on a set A is said to be an equivalence relation on A iff it is i) reflexive ii) symmetric iii) transitive

Note: i) The least equivalence relation on a given set A is the identity relation on A.

ii) The greatest equivalence relation is universal relation.

***W.E-5:* Let $R=\{(2,3)(3,4)\}$ be a relation defined on the set $A=\{1,2,3,4\}$ The minimum number of ordered pairs required to be added in R so that enlarged relation becomes an equivalance relation is**

Sol: Given $R=\{(2,3),(3,4)\}$

To make it reflexive,enlarge R as following

$R=\{(1,1),(2,2),(3,3),(4,4),(2,3),(3,4)\}$

Hence four more ordered pairs are added.

To make it symmetric, enlarge R as following

$R=\{(1,1),(2,2),(3,3),(4,4),$

$(2,3),(3,4),(3,2),(4,3)\}$

Hence two more ordered pairs are added.

Finally to make it transitive, we enlarge R to

$\{(1,1),(2,2),(3,3),(4,4),(2,3),(3,2),(3,4),$

$(4,3),(2,4)(4,2)\}$. Hence two more ordered pairs are added. $\therefore$ Total 8 ordered pair must be added to make the relation R an equivalence.

- **Ordered Relation:** A relation R is called ordered if R is transitive but not an equivalence relation.
- **Partial Order Relation:** A relation R is called partial order relation if R is reflexive, transitive and anti symmetric at the same time.

Eg : Let $A=\{1,2,3\}$ we defined

$R=\{(1,1),(2,2),(3,3)\}$ then R is both equivalence relation and partial order relation

- **Composition of relations :** Let R and S be two relations from sets A to B and B to C respectively. Then we can define a relation SoR from A to C such that

$(a,c)\in SoR \Leftrightarrow \exists b\in B$ such that

$(a,b)\in R$ and $(b,c)\in S$ .This relation is called the composition of R and S. In general

$RoS\neq SoR$. Also $(SoR)^{-1}=R^{-1}oS^{-1}$.

Eg : Let $A=\{1,2,3\}$ $B=\{x,y\}$ $C=\{a,b,c\}$

Let $R=\{(1,x),(1,y),(3,y)\}\subseteq A\times B$

$S=\{(x,a),(x,b),(y,b),(y,c)\}\subseteq B\times C$ Then

$SoR=\{(1,a),(1,b),(1,c),(3,b),(3,c)\}\subseteq A\times C$

because $(1,x)\in R$ and $(x,a)\in S\Rightarrow(1,a)\in SoR$

- **Congruence Modulo (m):** Let m be any positive integer. The integer 'a' is said to be congruent to 'b' of modulo 'm' if (a–b) is divisible by m. we write $a\equiv b(\bmod m)$ Thus

$a\equiv b(\bmod m)\Leftrightarrow(a-b)$ is divisible by m

Eg : i) $17\equiv 2(\bmod 5)$ as $17-2=15$ which is divisible by 5.

ii) $4\equiv 14(\bmod 5)\Rightarrow 4-14\equiv(\bmod 5)\Rightarrow -10$ is divisible by 5.

Note: Congruence modulo m is always an equivalence relation.

***W.E-6:* The congruent solution of $8x\equiv 6(\bmod 14)$ is**

(a) $x=6,7$ ***(b)*** $x=6,13$

(c) $x=2,13$ ***(d)*** $x=2,3$

Sol : Given , $8x\equiv 6(\bmod 14)$

$\therefore\ \lambda=\frac{8x-6}{14}$, where $\lambda\in I$

$\therefore\ 8x=14\lambda+6\Rightarrow x=\frac{14\lambda+6}{8}$

$\Rightarrow x=\frac{7\lambda+3}{4}=\frac{4\lambda+3(\lambda+1)}{4}$

$x=\lambda+\frac{3}{4}(\lambda+1)$, where $\lambda\in I$

and here greatest common divisor of 8 and 14 is 2. So, there are two required solution for $\lambda=3$ and $\lambda=7$ $\Rightarrow x=6,13$.

Hence, (b) is the correct answer.

W.E-7: The relation "congruence modulo m" on the set Z of all integer is an equivalence relation

Sol: Let $a \in I$ then $a - a = 0 = 0 \times m$

i) $\Rightarrow a - a$ is divisible by m, $\Rightarrow a \equiv a(\bmod m)$

$\Rightarrow R$ is reflexive

ii) $a, b \in Z$ such that $a \equiv b(\bmod m)$

as $a - b$ is divisible by m

$\Rightarrow a - b = \lambda m, \forall \lambda \in Z$

$\Rightarrow (b-a) \equiv (-\lambda)m \Rightarrow (b-a)$ is divisible by m.

$\Rightarrow b \equiv a(\bmod m)$ $\Rightarrow R$ is symmetric on Z

iii) Let $a, b, c \in Z$ such that

$a \equiv b(\bmod m).b \equiv c(\bmod m)$

$\therefore a \equiv b(\bmod m) \Rightarrow a - b$ is divisible by m

$\therefore a - b = \lambda_1 m$ for some

similarly $b - c = \lambda_2 m$ for some $\lambda_2 \in Z$

By (i) and (ii) we have

$a - c = (\lambda_1 + \lambda_2)m = km$ for some $k \in Z$

$\therefore a - c$ is divisible by m

$\therefore a \equiv c(\bmod m)$

$\therefore$ Congruence modulo m is transitive on Z

As the congruence modulo m is reflexive, symmetric, transitive so it is an equivalence relation on Z.

➢ **Some facts on relations :** Let A be a finite set and $n(A) = n$, then

(i) the number of elements in $A \times A$ is n^2

(ii) the number of relations from A to A is 2^{n^2}

(iii) number of reflexive relations from A to A is 2^{n^2-n}

(iv) number of symmetric relations from A to A is $2^{\frac{n^2+n}{2}}$

(v) number of relations from A to A which are not symmetric is $2^{n^2} - 2^{\frac{n(n+1)}{2}}$

(vi) number of relations from A to A which are both reflexive and symmetric is $2^{\frac{n^2-n}{2}}$

(vii) number of relations from A to A which are symmetric but not reflexive is $2^{\frac{n(n+1)}{2}} - 2^{n^2-n}$

➢ (i) Total number of relations from the set A to set B is $2^{n(A)n(B)}$

(ii) Let A and B be two non-empty sets having n elements in common then number of elements common in $(A \times B) \cap (B \times A) = n \times n = n^2$

➢ (i) Let A be a finite set. If $B_1, B_2, ...B_n$ are non-empty subsets of A such that $B_1 \cup B_2 \cup .... \cup B_n = A$ and $B_i \cap B_j = \phi$ for $i \neq j$ then $P = \{B_1, B_2, ..., B_n\}$ is called a partition of A.

Eg : Let $A = \{1,2,3,4,5,6,7\}$

suppose $B_1 = \{1,5\}, B_2 = \{2,4,7\}, B_3 = \{3,6\}$

and Clearly $B_1 \cup B_2 \cup B_3 = S$ and

$B_1 \cap B_2 = \phi,\ B_2 \cap B_3 = \phi$

$\therefore P = \{B_1, B_2, B_3\}$ is a partition of the set A.

➢ **Some facts on relations :**

(i) The intersection of two equivalence relations on a set A is an equivalence relation on set A.

(ii) Inverse of an equivalence relation is an equivalence relation.

(iii) The union of two equivalence relation on a set is not necesarily an equivalence relation on the set.

(iv) The identity relation on a non-empty set is always equivalence relation i.e. it is reflexive , symmetric as well as transitive.

(v) The identity relation on a set (non-empty) is always antisymmetric relation.

(vi) The universal relation on a non-empty set is always equivalence relation.

(vii) If R be a relation from as set X to Y and R be a relation Y to Z then $(RoS)^{-1} = S^{-1}oR^{-1}$

(viii) The number of equivalence relations on a finite set A is equal to number of partitions of A.

(ix) If R is an equivalence relation on a set A then R^{-1} is also an equivalence relation.

➢ Some Results on Cartesian Product of sets :

(i) $A\times(B\cup C)=(A\times B)\cup(A\times C)$

(ii) $A\times(B\cap C)=(A\times B)\cap(A\times C)$

(iii) $A\times(B-C)=(A\times B)-(A\times C)$

(iv) If A and B are two non-empty sets,then $A\times B=B\times A \Leftrightarrow A=B$

(v) If $A\subseteq B$, then $A\times A\subseteq(A\times B)\cap(B\times A)$

(vi) $A\subseteq B \Rightarrow A\times C\subseteq B\times C$ for any set C

(vii) $A\subseteq B$ and $C\subseteq D \Rightarrow A\times C\subseteq B\times D$

(viii) $(A\times B)\cup(C\times D)\subseteq(A\cup C)\times(B\cup D)$

(ix) $(A\times B)\cap(C\times D)=(A\cap C)\times(B\cap D)$

(x) $(A\times B)\cap(B\times A)=(A\cap B)\times(B\cap A)$

(xi) Let A and B be two non-empty sets having n elements in common,then $A\times B$ and $B\times A$ have n^2 elements in common.

C.U.Q

1. Let R be the relation over the set of all straight lines in a plane such that $l_1Rl_2 \Leftrightarrow l_1\perp l_2$. Then R is

1) symmetric 2) reflexive
3) transitive 4) an equivalence

2. The relation 'greater than' denoted by '>' in the set of integer is

1) reflexive 2) symmetric
3) transitive 4) equivalence

3. Let R be the realtion on the set of all lines in a plane defined by $(l_1,l_2)\in R\exists l_1 \parallel l_2$ then R is

1) reflexive 2) symmetric
3) transitive 4) equivalence

4. Let R be the relation 'is congruent to' on the set of all triangles in a plane is

1) reflexive 2) symmetric
3) transitive 4) equivalence

C.U.Q-KEY

01) 1	02) 3	03) 4	04) 4

C.U.Q-HINTS

1. Clearly, R is not a reflexive relation, because a line cannot be perpendicular to itself.
Let l_1Rl_2, Then $l_1Rl_2 \Rightarrow l_1\perp l_2 \Rightarrow l_2\perp l_1$
$\Rightarrow l_2Rl_1$
$\therefore$ R is a symmetric relation. R is not a transitive relation, because if $l_1\perp l_2$ and $l_2\perp l_3$, then l_1 is not perpendicular to l_3.

2. since $x>y, y>z \Rightarrow x>z \forall x,y,z\in N$
$xRy, yRz \Rightarrow xRz$

3. Define relation R on the set L of the all straight lines in a plane such that $(l_1,l_2)\in R \Leftrightarrow l_1 \parallel l_2$;
l_1 and l_2 are two straight lines is reflexive, for lRl for every straight line.
(every straight line parallel to itself)
Similarly R satisfies symmetric, transitive property

4. Let S denote the set of all triangles in a plane. Let R be the relation on S defined by
$(\Delta_1,\Delta_2)\in R \Leftrightarrow$ triangles $\Delta_1\cong\Delta_2$
i) Let each triangles $\Delta\in S$, we have $\Delta\cong\Delta$
$\Rightarrow (\Delta,\Delta)\in R \forall \Delta\in S \Rightarrow R$ is reflexive.
ii) $(\Delta_1,\Delta_2)\in R \Rightarrow \Delta_1\cong\Delta_2 \Rightarrow \Delta_2\cong\Delta_1$
$\Rightarrow (\Delta_2,\Delta_1)\in R \Rightarrow$ R is symmetric
Similarly we can show that R is transitive.

LEVEL - I (C.W)

Domain, Range and Number of Relations :

1. If P={a,b,c} and Q={r}, Then

1) P x Q $\neq$ Q x P 2) P x Q = Q x P
3) n (P x Q) $\neq$ n (Q x P) 4) 1 and 3

2. **Let A= {1,2,3} ,B={3,4} and C={4,5,6}.Then A x (B $\cap$ C) =.........**

1) {(1,4)} 2) {(2,4)} 3) {(2,4),(3,4)} 4) {(1,4),(2,4),(3,4)}

3. **The relation R defined on the set A={1,2,3,4,5} by R= { (a, b) : | a^2 - b^2| < 16; a,b $\in$ A} is given by**

1) { (1,1),(2,1),(3, 1),(4,1),(2,3) }
2) {(2, 2), (3, 2), (4, 2), (2,4)}
3) {(3, 3), (4, 3), (5, 4), (3, 4)}
4) {(1,1),(1,2),(1,3),(1,4),(2,1),(2,2), (2,3), (2,4),(3,1),(3.2),(3,3),(3,4),(4,1), (4,2), (4,3), (4,4),(4,5), (5,4),(5,5)}

4. **If the relation $R: A \to B$, where $A=\{1,2,3\}$ and $B=\{1,3,5\}$ is defined by $R=\{(x,y): x<y, x\in A, y\in B\}$, then**

1) $R=\{(1,3),(1,5),(2,3),(2,5),(3,5)\}$
2) $R=\{(1,1),(1,5),(2,3),(3,5)\}$
3) $R^{-1}=\{(3,1),(5,1),(3,2),(5,3)\}$
4) $R^{-1}=\{(1,1),(5,1),(3,2),(5,3)\}$

5. **Let Y = { 1,2,3,4,5}, A = {1,2}, B ={3,4,5} and ϕ denote the null set. If A $\times$ B denotes the cartesian product of sets A and B, then (Y $\times$ A) $\cap$ (Y $\times$ B) is**

1) Y 2) A 3) B 4) ϕ

6. **Given A={1,2,3,4,5,6}. Define a relation R from A to A by R={(x,y)/ $x, y \in A$; y=x+1}.Then domain of R is**

1) {1,2,3,4,5} 2) {2,3,4,5,6} 3) {1,2,3} 4) {4,5,6}

7. **If a relation 'R' is defined by $R=\{(x,y)/2x^2+3y^2\le 6\}$, then the domain of 'R' is**

1) $[-3,3]$ 2) $[-\sqrt{3},\sqrt{3}]$ 3) $[-\sqrt{2},\sqrt{2}]$ 4) $[-2,2]$

8. **Let A be a set of first ten natural numbers and R be a relation on A, defined by (x,y) $\in$ R $\Leftrightarrow$ x + 2y = 10,then domain of R is**

1) {1,2,3,......, 10} 2) {2,4,6,8} 3) {1,2,3,4} 4) {2,4,6,8,10}

9. **Let A={x,y,z} B={1,2}.Then the number of relations from A to B**

1) 2^6 2) 2^3 3) 2^2 4) 2^7

Types of Relations :

10. **Let R be a reflexive relation on a finite set A having n elements and let there be m ordered pairs in R, then**

1) $m \ge n$ 2) $m \le n$ 3) $m = n$ 4) $m < n$

11. **Let R be a reflexive relation on a set A and I be the identity relation on A. Then**

1) $R \subset I$ 2) $I \subseteq R$ 3) $R = I$ 4) all the above

12. **Let A be the set of the children in a family. The relation 'x is a brother of y' The relation on A is**

1) reflexive 2) symmetric 3) transitive 4) anti symmetric

13. **Let $A=\{1,2,3\}$ and $R=\{(1,1),(1,3),(3,1),(2,2),(2,1),(3,3)\}$, then the relation R on A is**

1) reflexive 2) symmetric 3) transitive 4) equivalence

14. **Let $A=\{2,4,6,8\}$ and $R=\{(2,4),(4,2),(4,6),(6,4)\}$ then R is**

1) reflexive 2) symmetric 3) transitive 4) anti symmetric

15. **Let A ={1,2,3,4}, R={(2,2),(3,3),(4,4),(1,2)} be a relation on A. Then R is**

1) reflexive 2) symmetric 3) transitive 4) Both 1 & 2

16. **N is the set of natural numbers. The relation R is defined on $N\times N$ as follows $(a,b)R(c,d) \Leftrightarrow a+d=b+c$. Then R is**

1) reflexive only 2) symmetric only 3) transitive only 4) an equivalence relation

17. **Let $R=\{(1,3),(4,2),(2,4),(2,3),(3,1)\}$ be a relation on the set $A=\{1,2,3,4\}$. Then the relation R is**

1) not symmetric 2) transitive 3) a function 4) reflexive

18. **In the set $A=\{1,2,3,4,5\}$, a relation R is defined by $R=\{(x,y):x,y\in A,x<y\}$. Then R is**
1) reflexive 2) symmetric
3) transitive 4) equivalence

19. **If $A=\{1,2,3\}$, the number of reflexive relation in A is**
1) 9 2) 3 3) 64 4) 68

20. **Let R and S be two non-void relations on set A which of the following statements is false.**
1) R and S are transitive $\Rightarrow R\cup S$ is transitive
2) R and S are transitive $\Rightarrow R\cap S$ is transitive
3) R and S are symmetric $\Rightarrow R\cup S$ is symmetric
4) R and S are symmetric $\Rightarrow R\cap S$ is symmetric

21. **Two points A and B in a plane are related if OA = OB, where O is a fixed point. This relation is**
1) partial order relation 2) equivalence relation
3) reflexive but not symmetric
4) reflexive but not transitive

22. **Which one of the following relations on Z is equivalence relation?**
1) $xR_1y\Leftrightarrow |x|=|y|$ 2) $xR_2y\Leftrightarrow x\ge y$
3) $xR_3y\Leftrightarrow x/y$ 4) $xR_4y\Leftrightarrow x<y$

Inverse Relation:

23. **If the relation $R:A\to B$, where $A=\{1,2,3,4\}$ and $B=\{1,3,5\}$ is defined by $R=\{(x,y):x<y,x\in A,y\in B\}$, then RoR^{-1} is**
1) $\{(1,3),(1,5),(2,3),(2,5),(3,5),(4,5)\}$
2) $\{(3,1),(5,1),(5,2),(5,3),(5,4)\}$
3) $\{(3,3),(3,5),(5,3),(5,5)\}$ 4) $\{(3,5)\}$

24. **R is a relation from {11,12,13} to {8,10,12} defined by y=x-3. Then R^{-1} is**
1) {(8,11),(10,13)}
2) {(11,8),(13,10)}
3) {(10,11),(8,11)}
4) {(11,8),(10,13),(12,15)}

LEVEL-I (C.W)-KEY

01) 1	02) 4	03) 4	04) 1
05) 4	06) 1	07) 2	08) 2
09) 1	10) 1	11) 2	12) 3
13) 1	14) 2	15) 3	16) 4
17) 1	18) 3	19) 3	20) 1
21) 2	22) 1	23) 3	24) 1

LEVEL - I (C.W) HINTS

1. $P\times Q=\{(a,r),(b,r),(c,r)\}$, $Q\times P=\{(r,a),(r,b),(r,c)\}$
2. $B\cap C=\{4\}$
3. We have $(a, b)\in R \Leftrightarrow |a^2 - b^2| < 16$.
$a = 1 \Rightarrow 0< b^2 < 17 \Rightarrow b= 1, 2, 3, 4$.
If a = 2 we get b= 1, 2, 3, 4.similary a =3 we get b =1,2,3,4,a=4 we get b=1,2,3,4,5,a=5 we get b=4,5
Then R={(1,1),(1,2),(1,3),(1,4),(2,1),(2,2),(2,3),(2,4),(3,1),(3.2),(3,3),(3,4),(4,1),(4,2),(4,3), (4,4),(4,5),(5,4),(5,5)}
Clearly, option (4) is correct.
4. Since, $R=\{(x,y):x<y,x\in A,y\in B\}$ i.e.R contains all those pairs for which the first number from the set A is less than the second from the set B.
5. $(Y\times A)\cap(Y\times B)=Y\times(A\cap B)=Y\times\phi=\phi$
6. $R=\{(1,2),(2,3),(3,4),(4,5),(5,6)\}$
7. $R=\{(x,y)/2x^2+3y^2\le 6\}$, then
$\frac{x^2}{3}+\frac{y^2}{2}\le 6\Rightarrow x\in[-\sqrt{3},\sqrt{3}]$
8. $x+2y=10\Rightarrow x=10-2y$
$A=\{1,2,3,\ldots\ldots\ldots\ldots 10\}$
Domain of $R=\{2,4,6,8\}$
$R=\{(2,4),(4,3),(6,2),(8,1)\}$
9. $2^{n(A)\times n(B)}=2^{3\times 2}=2^6$
10. The set consists of n elements and for relation to be reflexive it must have at least n ordered pairs. it has m ordered pairs therefore m $\ge$ n.
11. Each element of I is an element of R,
$n(I)\le n(R)$
12. Let R be the given relation. Let $(x_1,y_1)\in R$,

$(y_1, y_2) \in R \Rightarrow (x_1, y_2) \in R$

13. Reflexive but not Symmetric as $(2,1) \in R$ but $(1,2) \notin R$

Not Transitive As (2, 1) and $(1,3) \in R$ but $(2,3) \notin R$

14. $aRb \Rightarrow bRa$

15. Since $(1,1) \notin R$. So, R is not reflexive.
Now $(1,2) \in R$ but $(2,1) \notin R$,
$\therefore$ R is not symmetric.Clearly R is transitive.

16. We have $(a,b)R(a,b)$ for all $(a,b) \in N \times N$ since $a+b=b+a$.
Hence R is reflexive.R is symmetric
we have
$(a,b)R(c,d) \Rightarrow a+d=b+c \Rightarrow d+a=c+b$
$\Rightarrow c+b=d+a \Rightarrow (c,d)R(a,b)$ R is transitive
Let $(a,b)R(c,d)$ and $(c,d)R(e,f)$
Then by definition of R, we have $a+d=b+c$ and $c+f=d+e$ or $a+f=b+e$.
Hence $(a,b)R(e,f)$ Thus $(a,b)R(c,d)$ and $(c,d)R(e,f) \Rightarrow (a,b)R(e,f)$

17. $(3,2) \notin R$

18. $xRy, yRz \Rightarrow xRz$

19. 2^{n^2-n}

20. Let $A=\{1,2,3\}$ *and* $R=\{(1,1)(1,2)\}$ and $S=\{(2,2)(2,3)\}$ clearly R and S are transitive relations on A now
$R \cup S=\{(1,1)(2,2)(1,2)(2,3)\}, R \cup S$ is not transitive as $(1,3) \notin R \cup S$

21. $OA=OA, OA=OB \Rightarrow OB=OA$,
$OA=OB, OB=OC \Rightarrow OA=OC$ Hence R, S, T i.e. equivalence relation.

22. We observe that R_1 is reflexive, symmetric and transitive relation on Z.Relations R_2, R_3 and R_4 are not symmetric on Z.

23. $R=\{(1,3),(1,5),(2,3),(2,5),(3,5),(4,5)\}$

and

$R^{-1}=\{(3,1),(5,1),(3,2),(5,2),(5,3),(5,4)\}$

Thus $RoR^{-1}=\{(3,5),(5,5),(3,3),(5,3)\}$

24. $\because y=x-3, R=\{(11,8),(13,10)\}$,

$R^{-1}=\{(8,11),(10,13)\}$

LEVEL - I (H.W)

Domain, range and Number of Relations :

1. If P={1,2}, Then P x P x P is
1) {(1,1,1,),(1,1,2),(1,2,1),(1,2,2),(2,1,1),(2,1,2),(2,2,1),(2,2,2,)}
2) {(1,1,1),(1,2,2),(1,2,4)}
3) {(1,1,3)}
4) All the above

2. If $A=\{x: x^2-5x+6=0\}, B=\{2,4\}$, $C=\{4,5\}$, then $A \times (B \cap C)$ is
1) $\{(2,4),(3,4)\}$
2) $\{(4,2),(4,3)\}$
3) $\{(2,4),(3,4),(4,4)\}$
4) $\{(2,2),(3,3),(4,4),(5,5)\}$

3. If $A=\{x: x^2-5x+6=0\}, B=\{1,2\}$ and $C=\{4,5\}$ then $(A-B)\times(A-C)=$
1) $\{(2,3)\}$
2) $\{(1,2)\}$
3) $\{(1,2),(2,3)\}$
4) $\{(3,2)(3,3)\}$

4. A={1,2,3,4,}, relation R on A is defined by R={(x,y) / x<y and $|x^2-y^2|<9$;x,y$\in$A} then R=
1) {(1,1) (2,2) (3,3) (4,4)}
2) {(2,1) (3,2) (3,2) (4,3)}
3) {(1,2) (1,3) (2,3) (3,5)}
4) {(1,2) (1,3) (2,3) (3,4)}

5. **Let A={1,2,3,............,14}. Define a relation R from A to A by R={$(x, y): 3x - y = 0;\ x, y \in A$}.Then domain of R is**

1) {3,6,9,12} 2) {3,6}
3) {1,2,3,...........,14} 4) {1,2,3,4}

6. **The domain and range of relation** $R = \{(x, y) / x, y \in N, x + 2y = 5\}$ **is**

1) {1, 3}, {2, 1} 2) {2, 1}, {3, 2}
3) {1, 3}, {1, 1} 4) {1, 2}, {1, 3}

7. **If R ={(x,y): x,y $\in$ Z , $x^2 + y^2 \le 4$} is a relation in Z, then domain of R is**

1) {0,1,2} 2) {0,-1,-2}
3) {-2,-1,0,1,2} 4) {1,2,3}

8. **If R = {(x,y): x, y $\in$ N, y is the remainder when x is divided by 7}.Then sum of all numbers in range of R**

1) 14 2) 21 3) 28 4) 12

9. **Write the relation** $R = \{(x, x^3) : x$ **is prime number less than 10**} **in roaster form**

1) $R = \{(2,8),(3,27),(5,125),(7,343)\}$
2) $R = \{(2,4),(3,9),(5,25),(7,49)\}$
3) $R = \{(2,2),(3,3),(5,5),(7,7)\}$
4) $R = \{(2,8),(3,9),(5,25),(7,343)\}$

10. **A relation R is defined in the set of integers I as follows** $(x, y) \in R$ **iff** $x^2 + y^2 = 9$**, which of the following is true?**

1) $R = \{(0,3),(0,-3),(3,0),(-3,0)\}$
2) Domain of R = {–3, 0, 3}
3) Range of R = {–3, 0, 3} 4) All the above

Types of Relations :

11. **R = {(a,b): a,b $\in$ N, a+b is even} is**

1) reflexive 2) Symmetric
3) both 1,2 4) none of 1,2

12. **Let** $X = \{1, 2, 3\}$ **and** $R = \{(1,1),(2,2),(3,3),(2,3)\}$ **be a relation on X. Then which one is not true**

1) R is reflexive 2) R is transitive
3) R is antisymmetric 4) R is symmetric

13. **Let** $A = \{a, b, c\}$ **and** $R = \{(a,a),(b,b),(a,b),(b,a),(b,c)\}$ **be a relation on A, then R is**

1) reflexive 2) symmetric
3) transitive 4) not reflexive

14. **The relation R={(1,1)(2,2)(3,3)} on the set {1,2,3} is**

1) Symmetric only 2) Reflexive only
3) Transitive only 4) An equivalence

15. **Which of the following are not equivalence relations on I ?**

1) aRb if a + b is an even integer
2) aRb if a – b is an even integer
3) aRb if a < b 4) aRb if a = b

16. **Total number of equivalence relations defined in the set** $S = \{a, b, c\}$ **is**

1) 5 2) 3! 3) 2^3 4) 3^3

17. **Let** $R = \{(1,3),(4,2),(2,4),(2,3),(3,1)\}$ **be a relation on the set A = {1, 2, 3, 4}. The relation R is [AIE-2004]**

1) a function 2) transitive
3) not symmetric 4) reflexive

18. **Let R = {(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)} be a relation on the set A = {3, 6, 9, 12}. The relation is[AIE-2005]**

1) reflexive and symmetric only
2) an equivalence relation
3) reflexive only
4) reflexive and transitive only

19. **In the set Z of all integers, which of the following relation R is not an equivalence relation?**

1) x R y : if x $\le$ y 2) x R y : if x=y
3) x R y : if x-y is an even integer
4) x R y : if x $\equiv$ y (mod 3)

20. **Which of the following is an equivalence relation ?**

1) $x < y$ 2) $x > y$
3) $x - y$ is divisible by 5 4) x divides y

21. **Let W denote the words in the English dictionary. Define the relation R by** $R = \{(x, y) \in W \times W$ **/ the words *x* and *y* have atleast one letter in common}Then R is [AIE-2006]**

1) reflexive, symmetric and not transitive
2) reflexive, symmetric and transitive
3) reflexive, not symmetric and transitive
4) not reflexive, symmetric and transitive

Inverse Relation :

22. **If** $R=\{(x,y)\,|\,x\in N, y\in N, x+3y=12\}$

then R^{-1} **is**

1) {(2, 9),(2, 6),(3, 3)} 2) {(3, 1),(2, 4),(3, 6)}
3) {(3,3),(2,6),(1,9)} 4) {(1,3), (1,6), (1,9)}

LEVEL-I (H.W)-KEY

01) 1	02) 1	03) 4	04) 4
05) 4	06) 1	07) 3	08) 2
09) 1	10) 4	11) 3	12) 4
13) 4	14) 4	15) 3	16) 1
17) 3	18) 4	19) 1	20) 3
21) 1	22) 3		

LEVEL-I (H.W)-HINTS

1. $p\times p\times p=\{(1,1,1,),(1,1,2),$ $(1,2,1),(1,2,2),(2,1,1),$ $(2,1,2),(2,2,1),(2,2,2,)\}$

2. We have, $A=\{2,3\}$, $B=\{2,4\}$ and $C=\{4,5\}\ \therefore B\cap C=\{4\}$
$\Rightarrow A\times(B\cap C)=\{(2,4),(3,4)\}$

3. $x^2-5x+6=0\Rightarrow A=\{2,3\}$

4. Since x<y and $|x^2-y^2|<9$
$R=\{(1,2),(1,3),(2,3),(3,4)\}$ x<y

5. $\because 3x-y=0\Rightarrow y=3x$
$\therefore R=\{(1,3),(2,6),(3,9),(4,12)\}$

6. $x+2y=5\Rightarrow x\in N, y\in N$.
Domain = Set of values of x = {1, 3}
Range = Set of values of y = {1, 2}

7. $x^2+y^2\le 4$ represents all points interior to the circle $x^2+y^2=4$
hence $-2\le x\le 2$ *and* $-2\le y\le 2$ integral values of x *are* $-2,-1,0,1,2$

8. Range = {0,1,2,3,4,5,6}

9. $R=\{(2,8),(3,27),(5,125),(7,343)\}$

10. $y=\pm\sqrt{9-x^2}$
$\Rightarrow R=\{(0,3),(0,-3),(3,0),(-3,0)\}$

11. $a+b$ even if both a,b are even or both a,b are odd

12. $(3,2)\notin R$

13. $(c,c)\notin R$

14. It is an equivalence relation

15. The relations given in OPTIONS (1), (2) and (4) are clearly reflexive, symmetric and transitive On the other hand, the relation of OPTION (C) is transitive but neither reflexive nor symmetric.

16. The smallest equivalence relation is the identity relation $R_1=\{(a,a),(b,b),(c,c)\}$.
Then two ordered pairs of two distinct elements can be added to give three more equivalence relation
$R_2=\{(a,a),(b,b),(c,c),(a,b),(b,a)\}$
Similarly R_3 and R_4. Finally the largest equivalence relation i.e., the universal relation R_5 = {(a, a), (b,b), (c,c),(a, b),(b, a),(a, c),(c,a), (b,c), (c,b)}

17. $(2,4),(2,3)\in R\Rightarrow 2$ has two images
$\Rightarrow$ R is not a reflexive
$(1,1)\notin R\Rightarrow R$ is not reflexive
$(2,3)\in R,(3,2)\notin R\Rightarrow R$ is not symmetric

18. $A=\{3,6,9,12\}$ and $R=\{(3,3),(6,6),(9,9),(12,12)\}\in R\Rightarrow$ R is reflexive $(6,12)\in R$ but $(12,6)\notin R\Rightarrow$ R is not symmetric
$(3,6)\in R,(6,12)\in R\Rightarrow(3,12)\in R$
$\therefore$ R is transitive

19. If R is relation defined by x R y : then R is not an equivalence relation.

20. xRx since ; 0 is divisible by 5 if xRy, x-y is divisible by 5, y-x is also divisible by 5 if xRy & yRz, $x-y=5k_1$; $y-z=5k_2$; $x-z=5(k)$

21. Let $W=\{CAT, TOY, YOU.......\}$
clearly R is reflexive and symmetric but not transitive (CAT R TOY, TOY R YOU does not implies CAT R YOU.

22. $x+3y=12x=3, y=3$

$x=6, y=2$ $x=9$

$y=1$ $R=\{(3,3),(6,2),(9,1)\}$

$R^{-1}=\{(3,3),(2,6),(1,9)\}$

LEVEL - II (C.W)

Domain, Range and Number of Relations :

1 **If $A=\{(1,2)\}, B=\{(3,4)\}$, then $A\times(B\times\phi)=$**

1) ϕ 2) A 3) B 4) $\{1,3\}$

2. **R, S are relations from N × N to Z × Z by $R=\{(x-y, y-x): x, y\in N\}$, $S=\{(x-y, x+y): x, y\in Z\}$, Then number of elements in R ∩ S**

1) 0 2) 1 3) 2 4) infinite

3. **A={1,2,3,5}, B={4,6,9}. Define a relation R from A to B by R={(x,y);the difference between x and y is odd; x∈A, y∈B}. Then R is**

1) {(1,2),(1,6),(2,9),(3,4),(3,5)}
2) {(1,4),(1,6),(2,9),(3,4),(3,6),(5,4),(5,6)}
3) {(1,6),(1,7),(1,8),(2,9),(2,4),(2,9)}
4) {(1,5),(1,6),(1,7),(6,4),(6,9),(6,2)}

4. **Given A={1,2,3,4,5}, S={(x,y): x∈A, y∈A}. Then the ordered pair which satisfy x+y>8 then R is ....**

1) {(5,5) (5,3)} 2) {(4,5) (5,4) (5,5)}
3) {(5,4) (5,6)} 4) {(5,4) (5,3)}

5. **The domain and range of**

$$R=\{(x,y)/y=x+\frac{6}{x}, x, y\in N \text{ and } x<6\}$$

1) Domain $=\{1,2\}$, Range $=\{7,5\}$
2) Domain $=\{1,2,3\}$, Range $=\{7,5\}$
3) Domain $=\{1\}$, Range $=\{7,5\}$
4) Domain $=\{1,2\}$, Range $=\{7\}$

6. **If the number of relations on a finite set A having 'n' elements is 2^{16}, then 'n' equal to**

1) 15 2) 17 3) 4 4) 8

7. **If a relation R is defined on the set Z of integers as follows $(x,y)\in R \Leftrightarrow x^2+y^2=25$. Then domain of R =**

1) $\{3,4,5\}$ 2) $\{0,3,4,5\}$
3) $\{0,\pm3,\pm4,\pm5\}$ 4) $\{0,\pm5\}$

Types of Relations :

8. **Let R be the relation on the set of all real numbers defined by a R b if $|a-b|\le1$. Then R is**

1) reflexive and symmetric
2) symmetric only 3) transitive only
4) anti - symmetric only

9. **Let $R=\{(x,y): x, y\in A; x+y=5\}$ where A = {1,2,3,4,5} then**

1) R is not reflexive, symmetric and not transitive
2) R is an equivalence relation
3) R is reflexive, symmetric but not transitive
4) R is not reflexive, not symmetric but transitive

10. **On the set of natural numbers N, the relation R is defined by xRy iff x + y = 100 is**

1) reflexive 2) not reflexive
3) equivalence 4) not symmetric

11. **On the set of all vectors in space the relation R is defined by $\bar{a}R\bar{b} \Leftrightarrow \bar{a}.\bar{b}$ is scalar is**

1) symmetric 2) not symmetric
3) not reflexive 4) both 2 and 3

12. **If A={ 1, 2,3} Then a relation reflexive but not Symmetric on A is**

1) {(1,1), (1,2)} 2) {(1,1),(1,2),(2,1),(2,2)}
3) {(1,1),(2,2),(3,3)}
4) {(1,1),(2,2),(3,3),(2,3)}

13. **The correct statement of the following is**

1) The relation 'less than' on Z is anti-symmetric
2) The relation 'sister of' on the members of a family is transitive
3) the relation 'relatively prime' on N is reflexive
4) The relation 'perpendicular' on a set of lines in a plane is transitive

14. **If A ={ 1,2,3} ,R ={ (1,2),(1,1),(2,3)} Then minimum number of elements may be adjoined with the elements of R so that it may become transitive is**

1) 0 2) 1 3) 2 4) 3

Inverse Relation :

15. **If R is a relation is "greater than or equal**

to” from A ={1,2,3,4} to B ={4,5,6}, then R^{-1} =

1) {(4,4)} 2) ϕ 3) A x B 4) R

LEVEL - II (C.W)-KEY

01) 1	02) 1	03) 2	04) 2
05) 2	06) 3	07) 3	08) 1
09) 1	10) 2	11) 1	12) 4
13) 2	14) 2	15) 1	

LEVEL - II (C.W)-HINTS

1. $B\times\phi=\phi\Rightarrow A\times(B\times\phi)=\phi$
2. y - x = x+ y only if x = 0 But $\notin$ N
3. both x and y are odd
4. 4+5=9>8, 5+4=9>8, 5+5=10>8
5. $R=\{(1,7),(2,5),(3,5)\}$
6. $n^2=16$
7. $x^2+y^2=25$ represents all points on the circle $x^2+y^2=25$ Hence $-5\le x\le 5,-5\le y\le 5$. $\therefore$ integral values of x are $0,\pm3,\pm4,\pm5$.
8. $0\le 1\Rightarrow aRa$ $|a-b|\le 1\Rightarrow|b-a|\le 1$
9. R={(1,4),(4,1),(2,3),(3,2)}
10. $(1,1)\notin R$
11. $\bar{a}.\bar{b}=\bar{b}.\bar{a}$
12. $\{(1,1),(2,2),(3,3)(2,3)\}$ is reflexive not symmetric as $(2,3)\in R$ but $(3,2)\notin R$
13. xRy, yRz $\Rightarrow$ xRz
14. (1,3) need to be adjoined to make the relation transitive
15. $R=\{(4,4)\}$

LEVEL - II (H.W)

Domain, Range and Number of Relations :

1. If P={x : x<3 ,x $\in$ N}, Q={x :x $\le$3, x$\in$ W}.then ($p\cup Q$)x($p\cap Q$). where W is the set of whole numbers

1) $\{(0,1),(0,2),(1,1),(1,2),(2,1),(2,2),(0,0),(3,3)\}$
2) $\{(0,1),(0,2),(1,1),(1,2),(2,1),(2,2),(3,1),(3,2)\}$
3) $\{(0,1),(0,2),(1,1),(1,2),(2,1),(2,2),(0,3),(3,3)\}$
4) $\{(0,1),(0,2),(1,1),(1,2),(2,1),(2,2),(0,3),(0,0)\}$

2. The domain of the relation R defined by R={($x,x+5$): $x\in$ {0,1,2,3,4,5}}is

1) {1,2,3,4,5} 2) {0,1,2,6,7}
3) {5,6,7,8,9,10} 4) {0,1,2,3,4,5}

3. Range of R ={(x , y):x,y $\in$ z, x+3y=12}

1) $\{\pm12,\pm9,\pm6,\pm3,0\}$ 2) $\{0,\pm3,\pm6,..\}$
3) Z 4) N

4. The relation R is defined by R={(a,b):b=$|a+1|$ and $|a|\le 3,a,b\in Z$ } then range of R =

1) {-3,-2,-1,0,1,2} 2) {0,1,2,3,4}
3) {-3,-2,-1} 4) {0,-1,-2,-3}

Types of Relations :

5. Let R be an equivalence relation defined on a set containing 6 elements. Then the minimum numbers of ordered pairs that R should contain

1) 6 2) 12 3) 6^6 4) 36

6. R ={(a,b): a,b $\in$ R, a^2+b^2 =1 } is

1) reflexive 2) Symmetric
3) transitive 4) anti symmetric

7. { (x,y): x,y $\in$ Z, x-y is divisible by 5} is

1) reflexive 2) Symmetric
3) both 1,2 4) not reflexive

8. Let R be a relation on the set N defined by $\{(x,y):x,y\in N,2x+y=41\}$, then R is

1) reflexive 2) symmetric
3) transitive 4) not symmetric

9. Let R_1 be a relation defined in the set of real numbers by a R_1 b $\Leftrightarrow$ 1+ab >0, Then R_1 is

1) equivalence relation 2) transitive
3) symmetric 4) anti symmetric

10. If R is an equivalence relation on a set A, then R^{-1} is

1) reflexive only 2) symmetric but not transitive
3) equivalence 4) transitive

11. Let a relation R on N as $(a,b)\in R$ if H.C.F $(a,b)\neq 1$, then which of the following

statements is not true about R

1) reflexive 2) symmetric
3) antisymmetric 4) transitive

12. Let N denote the set of all natural numbers and R a relation on $N \times N$. Which of the following is an equivalence relation?

1) $(a,b)R(c,d)$ if $ad(b+c)=bc(a+d)$

2) $(a,b)R(c,d)$ if $a+d=b+c$

3) $(a,b)R(c,d)$ if $ad=bc$

4) All the above

LEVEL-II (H.W) KEY

01) 2	02) 4	03) 3	04) 2
05) 1	06) 2	07) 3	08) 4
09) 3	10) 3	11) 4	12) 4

LEVEL-II (H.W)-HINTS

Domain, Range and Number of Relations :

1. P={1,2} Q={0,1,2,3}

$P \cup Q = \{0,1,2,3\}, P \cap Q = \{1,2\}$

$\therefore (P \cup Q) \times (P \cap Q) = \{0,1,2,3\} \times \{1,2\}$

$\{(0,1),(0,2),(1,1),(1,2),(2,1),(2,2),(0,3),(0,0)\}$

2. Domain = x

3. For every $y \in Z$ we have $y \in Z \ni x+3y=12$

4. R={(3,2),(2,1),(1,0),(0,1),(1,2),(2,3),(3,4)}

5. The minimum number of ordered paris that R should contain 6 elements

6. $a^2+b^2=1 \Rightarrow b^2+a^2=1$

7. Clearly R is Reflexive , x - y is divisible by 5 $\Rightarrow$ y - x is also divisible by 5 so it is symmetric

8. We have R = {(1, 39), (2, 37), (3, 35), (4, 33), (5, 31), (6, 29), (7, 27), (8, 25), (9, 23), (10,21), (11, 19), (12, 17), (13, 15), (14, 13), (15,11), (16, 9), (17, 7), (18, 5), (19, 3), (20, 1)}

Since (1, 39) $\in R$, but (39, 1) $\notin R$

Therefore, R is not symmetric.

Clearly R is not reflexive.

Now, $(15,11) \in R$ and $(11,19) \in R$ but

$(15,19) \notin R$ So R is not transitive.

9. Let $a \in R \Rightarrow 1+a.a = 1+a^2 > 0 \Rightarrow (a,a) \in R_1$

$\therefore$ R_1 is reflexive on R.

Let $(a,b) \in R_1. \Rightarrow 1+ab > 0 \Rightarrow 1+ba > 0$

$\Rightarrow (b,a) \in R_1 \ \therefore R_1$ is symmetric on R.

Since $\left(1, \frac{1}{2}\right) \in R_1$ and $\left(\frac{1}{2}, -1\right) \in R_1$ but

$(1,-1) \notin R_1 \ \therefore \ R_1$ is not transitive on R.

10. for $(x,x) \in R \Rightarrow (x,x) \in R^{-1}$

Let $(x,y) \in R^{-1}, (y,x) \in R \Rightarrow (x,y) \in R$

$\Rightarrow (y,x) \in R^{-1}$

Let $(x,y),(y,z) \in R^{-1}$ $(y,x) \in R$ and

$(z,y) \in R$ $(z,x) \in R \Rightarrow (x,z) \in R^{-1}$

11. $(a,b) \neq 1, (b,c) \neq 1 \Rightarrow (a,c) = 1$

12. The relation in (1) is reflexive because $ab = ba$ and $a+b=b+a$, so that

$ab(b+a) = ba(a+b)$. i.e. $(a,b)R(a,b)$.

It is also symmetric because

$ad(b+c) = bc(a+d)$, i.e. $(a,b)R(c,d)$,

implies $cb(d+a) = da(c+b)$,

i.e. $(c,d)R(a,b)$. Simple computations show that relation (A) is Transitive, too, and that the relations in (B) and (C) are also reflexive, symmetric and transitive.

LEVEL - III

1. Let A be the set of first 10 natural numbers and let

$R = \{(x,y) / x \in A, y \in N \ and \ x+2y=10\}$

then $n\{dom(R^{-1})\} =$

1) 4 2) 5 3) 8 4) 10

2. Let $X = \{1,2,3,4,5\}$, the number of different ordered pairs (Y,Z) that can be formed such that $Y \subseteq X$, $Z \subseteq X$ and $Y \cap Z$ is empty, is

1) 3^5 2) 2^5 3) 5^3 4) 5^2

[AIE-2012]

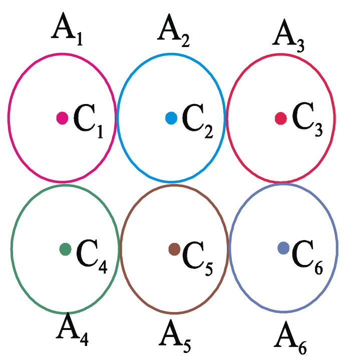
3. **The relation R defined on $A = \{1,2,3\}$ by aRb. if $|a^2 - b^2| \le 5$, which of the following is false.**

1) $R = \{(1,1),(2,2),(3,3),(2,1),(1,2),(2,3),(3,2)\}$
2) $R^{-1} = R$ 3) Domain of R = {1, 2, 3}
4) Range of R = {5}

4. **Let R be a relation such that R={(1,4),(3,7),(4,5),(4,6),(7,6)}, then $\left(R^{-1} o\, R\right)^{-1} =$**

1) {(1,1),(3,3),(4,4),(7,7),(4,7),(7,4),(4,3)}
2) {(1,1),(3,3), (4,4),(7,7)(4,7),(7,4)}
3) {(1,1),(3,3),(4,4)} 4) ϕ

5. **Let $A = \{A_1, A_2, A_3, A_4, A_5, A_6\}$ be the set of six unit circles with centres $C_1, C_2, C_3 ... C_6$ arranged as shown in the diagram.The relation R on A is defined by $\left(A_i, A_j\right) \in R \Leftrightarrow C_iC_j \le 2\sqrt{2}$ then**



1) R is symmetric, transitive but not reflexive
2) R is only transitive
3) R is symmetric, reflexive but not transitive
4) R is neither reflexive nor transitive but is symmetric.

6. **The relation R on the set of natural numbers N is defined as $xRy \Leftrightarrow x^2 - 4xy + 3y^2 = 0$, $x, y \in N$ then R is**

1) reflexive but not symmetric and not transitive
2) symmetric but not reflexive and not transitive
3) transitive but not reflexive and not symmetric
4) equivalence relation

7. **Which of the following relations is not transitive**

1) $(a,b) \in R_1 \Leftrightarrow a \le b, a, b \in z$

2) $(x,y) \in R_2 \Leftrightarrow x$ divides y if $x, y \in z$

3) $(x,y) \in R_3 \Leftrightarrow |x| + |y| = 1$, for $x, y \in R$

4) $(x_1, y_1) \in R_4 \Leftrightarrow l_1$ parallel to l_2.

where l_1, l_2 are lines

8. **A relation R on the set of non zero complex numbers is defined by $z_1 R z_2 \Leftrightarrow \dfrac{z_1 - z_2}{z_1 + z_2}$ is real, then R is**

1) Reflexive 2) Symmetric
3) Transitive 4) Equivalence

9. **S is a relation over the set R of all real numbers and it is given by (a, b) $\in$ S $\Leftrightarrow$ ab $\ge$ 0. Then S is**

1) symmetric and transitive only
2) reflexive and symmetric only
3) a partial relation 4) an equivalance relation

10. **Let R be the real line. consider the following subsets of the plane $R \times R$.**

$S = \{(x,y) : y = x+1$ **and** $0 < x < 2\}$,

T = {(x,y) : x-y is an integer}.Which one of the following is true ? [AIE-2008]

1) S is an equivalence relation on R but T is not
2) T is an equivalence relation on R but S is not
3) Neither S nor T is an equivalence relation on R
4) Both S and T are equivalence relations on R

11. **Consider the following relations**

R = { (x, y) / x, y are real numbers and x = wy for some rational number w};

S= $\left\{\left(\dfrac{m}{n}, \dfrac{p}{q}\right) / m, n, p \text{ and } q\right.$ **are integers such**

that $n, q \neq 0$ and qm=pn $\}$,then [AIE-2010]

1) neither R or S is an equivalence relation.
2) S is an equivalence relation but R is not an equivalence relation
3) R and S both are equivalence relations.
4) R is an equivalence relation but S is not an equivalence relation.

LEVEL-III-KEY

01) 1	02) 1	03) 4	04) 2
05) 3	06) 1	07) 3	08) 4
09) 4	10) 2	11) 2	

LEVEL-III-HINTS

1. $R=\{(2,4),(4,3),(6,2),(8,1)\}$

2. For any element x_i present in X, 4 cases arises while making subsets Y and Z.
Case 1 : $x_i \in Y, x_i \in Z \Rightarrow Y \cap Z \neq \phi$
Case 2 : $x_i \in Y, x_i \notin Z \Rightarrow Y \cap Z = \phi$
Case 3 : $x_i \notin Y, x_i \in Z \Rightarrow Y \cap Z = \phi$
Case 4 : $x_i \notin Y, x_i \notin Z \Rightarrow Y \cap Z = \phi$
For every element, number of ways=3 for which $Y \cap Z = \phi \Rightarrow$ total number of ways=$3\times3\times3\times3\times3$ [$\because$ no.of elements in set X=5]=$(3)^5$

3. $R=\{(1,1),(1,2),(2,1),(2,2),(2,3),(3,2),(3,3)\}$

4. R={(1,4),(3,7),(4,5),(4,6),(7,6)}
R^{-1}={(4,1),(7,3),(5,4),(6,4),(6,7)}

5. $(C_i, C_i) = 0 \leq 2\sqrt{2}$ R is reflexive
$(A_i, A_j) \in R \Rightarrow C_iC_j \leq 2\sqrt{2} \Rightarrow C_jC_i \leq 2\sqrt{2}$
$\therefore (A_j, A_i) \in R$
R is symmetric
if $(A_1, A_2) \in R, (A_2, A_3) \in R \Rightarrow (A_1, A_3) \notin R$

6. Given xRy iff $x^2 - 4xy + 3y^2 = 0$
If $y = x$, then $x^2 - 4x^2 + 3x^2 = 0$
Hence xRx $\forall x \in N$
$\therefore$ R is reflexive,
R is not symmetric, because $3 R 1 \not\Rightarrow 1R3$
R is not transitive,because.9R3 and 3R1 $\not\Rightarrow$ 9R1

7. $(0.6,0.4) \in R_3, (0.4,-0.6) \in R_3$ but $(0.6,-0.6) \notin R_3$

8. i) $z_1Rz_1 \Rightarrow \frac{z_1 - z_1}{z_1 + z_1} \forall z_1 \in C \Rightarrow 0$ is real
$\therefore$ R is a reflexive.
ii) $z_1Rz_2 \Rightarrow \frac{z_1 - z_2}{z_1 + z_2}$ is real
$\Rightarrow -\left(\frac{z_2 - z_1}{z_2 + z_2}\right)$ is real $\Rightarrow \left(\frac{z_2 - z_1}{z_2 + z_1}\right)$ is real
$\Rightarrow z_2Rz_1 \forall z_1, z_2 \in C \therefore$ R is a symmatric
iii) let $z_1 = a_1 + ib_1, z_2 = a_2 + ib_2$ and $z_3 = a_3 + ib_3$
when $a_1, b_1, a_2, b_2, a_3 b_3 \in R$
now $z_1Rz_2 \Rightarrow \frac{z_1 - z_2}{z_1 + z_2}$ is real
$\Rightarrow \frac{(a_1 - a_2) + i(b_1 - b_2)}{(a_1 + a_2) + i(b_1 + b_2)} \times \frac{(a_1 + a_2) - i(b_1 + b_2)}{(a_1 + a_2) - i(b_1 + b_2)}$
is real
$\Rightarrow (a_1 + a_2)(b_1 - b_2) - (a_1 - a_2)(b_1 + b_2) = 0$
(for purely real, imaginary part = 0)
$\Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2}$,similarly $z_2Rz_3 \Rightarrow \frac{a_2}{b_2} = \frac{a_3}{b_3}$
z_1Rz_2 and $z_2Rz_3 \Rightarrow \frac{a_1}{b_1} = \frac{a_2}{b_2}$ and $\frac{a_2}{b_2} = \frac{a_3}{b_3}$
$\Rightarrow \frac{a_1}{b_1} = \frac{a_3}{b_3}$ z_1Rz_3
R is transitive.
hence R is an equivalence relation.

9. Reflexivity : For any a $\in$ R, we have
$a^2 =$ aa $\geq 0 \Rightarrow$ (a,a) $\in$ S
Thus, (a,a) $\in$ s for all a $\in$ R.
So, S is a reflexive relation on R.
Symmetry: Let (a, b) $\in$ S. Then
(a, b) $\in$ S $\Rightarrow$ ab $\geq 0 \Rightarrow$ ba $\geq 0 \Rightarrow$ (b,a) $\in$ S
Thus (a, b) $\in$ S $\Rightarrow$ (b,a) $\in$ S for all a,b $\in$ R.
So S is symmetric relation on R.
Transitivity: Let a, b, c $\in$ S. such that
(a, b) $\in$ S and (b, c) $\in$ R $\Rightarrow$ ab $\geq$ 0 and

bc $\geq 0 \Rightarrow$ a, b, c are of same sign.

$\Rightarrow$ ac $\geq 0 \Rightarrow$ (a, c) $\in$ R.

(a, b) $\in$ S $\Rightarrow$ (b, c) $\in$ S $\Rightarrow$ (a, c) $\in$ S.

So, S is a transitive relation on R.

Hence, S is an equivalence relation on R.

10. $S=\{(x,y); y=x+1, 0<x<2\} \Rightarrow$

S is not symmetric

$T=\{(x,y): x-y \text{ is an integer}\}$

$\Rightarrow$ clearly T is an equivalence

11. xRy need not implies yRx, then clearly

S: $\frac{m}{n} s \frac{p}{q} \Leftrightarrow qm = pn$, $\frac{m}{n} s \frac{m}{n}$ reflexive

$\frac{m}{n} s \frac{p}{q} \Rightarrow \frac{p}{q} s \frac{m}{n}$ symmetric

$\frac{m}{n} s \frac{p}{q}, \frac{p}{q} s \frac{r}{s} \Rightarrow qm = pn, ps = rq$

$\Rightarrow ms = rn$ transitive

s is an equivalence relation

LEVEL - IV

1. **If R is a relation and R^{-1} is its inverse relation then**

statement I: Domain of R^{-1}=Range of R

statement II: Range of R^{-1}=Domain of R

1) I,II are true 2) I only true
3) II only true 4) Both I & II are false

2. **Consider the following relation R on the set of real square matrices of order 3.**

$R=\{(A,B) \mid A=P^{-1}BP$ for some invertible matrix $P\}$

Statement-I : R is equivalence relation.

Statement-II: For any two invertible 3×3 matrices M and N, $(MN)^{-1}=N^{-1}M^{-1}$

[AIE-2011]

1) Statement I is true, Statement II is a correct explanation for statement I
2) Statement I is true, statement II is true,Statement II is not a correct explanation for statement I
3) Statement I is true, statement II is false.
4) Statements I is false, statement II is true

3. **Let R be the set of real numbers**

Statement-1 :

$A=\{(x,y)\in R\times R: y-x \text{ is an integer}\}$

is an equivalence relation on R.

Statement-2: $B=\{(x,y)\in R\times R: x=\alpha y$ for some rational number $\alpha\}$ **is an equivalence relation on R. [AIEEE-2011]**

1) statement-1 is false, stattement - 2 is true.
2) statement-1 is true, stattement-2 is true; statement-2 is a correct explanation for statement-1.
3) statement-1 is true,statement-2 is true, statement-2 is not a correct explanation for statement-1.
4) statement-1 is true, statement-2 is false.

LEVEL - IV-KEY

01) 1	02) 2	03) 4

LEVEL - IV-HINTS

1. Domain of R^{-1} = Range of R

Range of R^{-1} = Domain of R

2. For statement I:

(i) Reflexive: $ARA \Rightarrow A=P^{-1}AP$ for some invertible matrix P.

Let $P=I$ then $ARA \Rightarrow A=I^{-1}AI$

$\therefore R$ is reflexive

(ii) Symmetric: $ARB \Rightarrow A=P^{-1}BP$

$\Rightarrow PAP^{-1}=P(P^{-1}BP)P^{-1}=B$

Since, for some invertible matrix P, we let

$Q=P^{-1}$, $\therefore B=(P^{-1})^{-1}AP^{-1} \Rightarrow B=Q^{-1}AQ$

$\Rightarrow BRA$ $\therefore$ R is symmetric

(iii) Transitive:

ARB and $BRC \Rightarrow A=P^{-1}BP$ and $B=P^{-1}CP$

$\Rightarrow A=P^{-1}(P^{-1}CP)P \Rightarrow A=(P^{-1})^2 C P^2$

so, ARC, for some $P^2=P \Rightarrow R$ is transitive

3. (A) $x-x=0\in I$ (it is reflexive)

$x-y\in I, y-x\in I$ (symmetric)

$x-y=I_1, y-z=I_2, x-z=I_3$

(transitive) (B) x R y need not implies y R x.